

Multiplication

* Multiplication is repeated addition.

$$\begin{array}{r} 1 \ 2 \ 3 \ 4 \\ + \ 5 \ 6 \ 7 \ 8 \\ \hline 6 \ 9 \ 1 \ 2 \\ \hline \end{array}$$

* Adding two n -digit numbers gives either an n -digit or $(n+1)$ -digit number.

~~* Running time = $O(n)$.~~

~~* Suppose two ^{positive integer} numbers a and b have the same number of digits. What is the time complexity of finding the sum of two numbers a and b ?~~

$$O(\max \{ \log_2(a), \log_2(b) \})$$

* Each digit takes at most two additions, so total time $\leq 2(n+1) = O(n)$.

* Subtraction is same as addition.

①.

	1	2	3	4	
		5	6	7	
	8	6	3	8	
7	4	0	4	0	
6	1	7	0	0	0
6	9	9	6	7	8

* Total multiplication performed
 $= 4 + 4 + 4 = 4 \times 3 = 12.$ $T(n) = n^2$

* Total multiplication performed
 with two n -digit numbers: $O(n^2).$ ↗

* Maybe there is scope of improvement! ②.

* 23-year-old Anatoly Karatsuba.

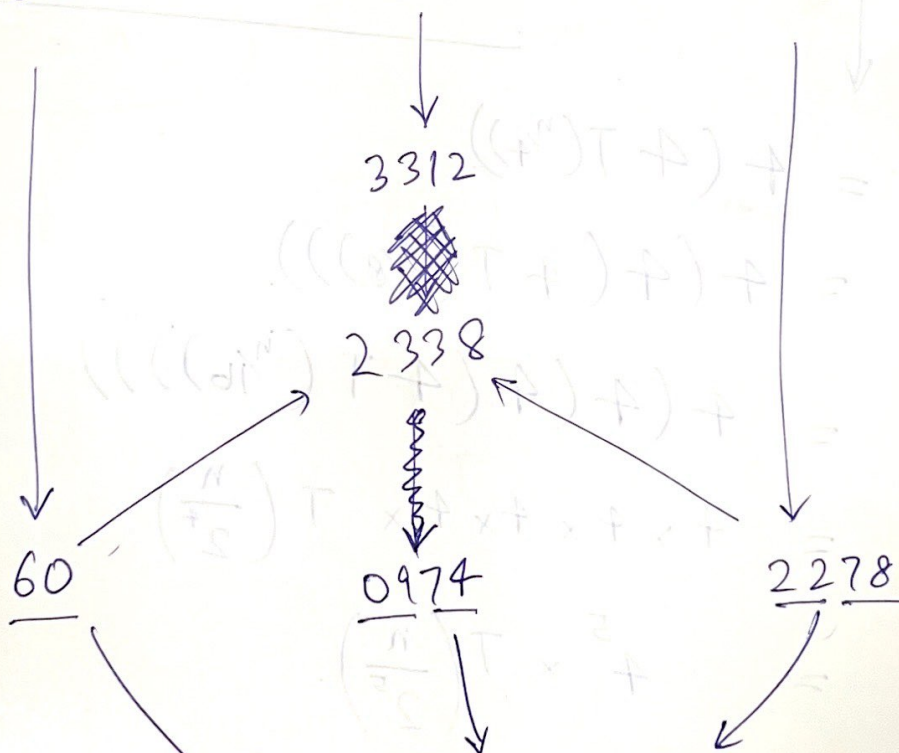
* Using high school tricks; no sophisticated mathematics involved!

Karatsuba's Algorithm.

[Horizontal arrows: addition,
vertical arrows: multiplication]

$$12 \longrightarrow 46 \longleftarrow 34$$

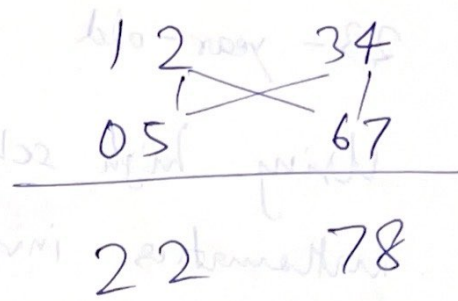
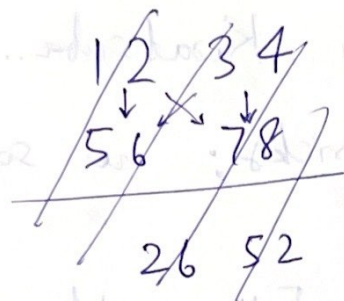
$$05 \longrightarrow 72 \longleftarrow 67$$



$$\boxed{\begin{array}{r} 69 \\ \hline 96 \\ \hline 78 \end{array}}$$

Ex. $\boxed{7145 \times 2698} = \begin{array}{r} 1927 \\ 7210 \end{array}$

(3)



$$T(n) = n^2.$$

$$T(n) = 4 T(n/2).$$



$$\begin{array}{r} 08 \quad 04 \quad 00 \\ 01 \quad 70 \quad 00 \\ 60 \quad 00 \quad 00 \\ \hline 69 \quad 96 \quad 78 \end{array}$$

$$\begin{aligned} T(n) &= 4(4 T(n/4)) \\ &= 4(4(4 T(n/8))) \\ &= 4(4(4(4 T(n/16)))) \\ &= 4 \times 4 \times 4 \times 4 \times T\left(\frac{n}{2^4}\right) \\ &= 4^5 \times T\left(\frac{n}{2^5}\right) \\ &= 4^5 \times \left(4 \times T\left(\frac{n}{2^6}\right)\right) \\ &= 4^6 \times T\left(\frac{n}{2^6}\right) = 4^k \times T\left(\frac{n}{2^k}\right) \quad (4) \end{aligned}$$

When is $\frac{n}{2^k} = 1$?

$$2^k = n \quad k = \log_2 n.$$

$$T(n) = 4^k \times T(1)$$

$$= 4^k = 4^{\log_2 n}$$

$$= n^{\log_2 4} = n^2$$

$$T(n) = n^2.$$

$$T(1) = 1$$

$$\begin{aligned} a^{\log_2 b} \\ = b^{\log_2 a} \end{aligned}$$

$$T(n) = 9 T\left(\frac{n}{3}\right)$$

$$= 9 \times 9 T\left(\frac{n}{9}\right)$$

$$= 9 \times 9 \times 9 T\left(\frac{n}{27}\right)$$

$$= 9^k T\left(\frac{n}{3^k}\right)$$

When is $\frac{n}{3^k} = 1$?

$$3^k = n$$

$$k = \log_3 n$$

(5)

- Research projects
- Patents
- Starting own company
- Consulting positions
- Industry collaborations.

$$T(n) = 9^k \times T(1)$$

$$= 9^{\log_3 n} \times 1$$

$$= n^{\log_3 9} = n^2$$

$$T(n) = n^2$$

$$T(n) = 3T\left(\frac{n}{2}\right)$$

Why does it work?

ab x cd.

$$(10^2a + b) \times (10^2c + d)$$

$$10^4ac + 10^3ad + 10^2bc + bd$$

$$a \times c \text{ (+00)} + a \times d \text{ (+0)} + b \times c \text{ (+0)} + b \times d$$

4 multiplications

(6)

$$10^4 ac + 10^2 ad + 10^2 bc + bd.$$

$$10^4 ac + 10^2(ad + bc) + bd.$$

Puzzle.

Use axc and bxd to compute $ad + bc$ with only one extra multiplication.

Given: $ac, bd.$

Compute: $ad + bc$ using one ~~extra~~ multiplication.

$$(a+b)(c+d) = ac + bd + ad + bc.$$

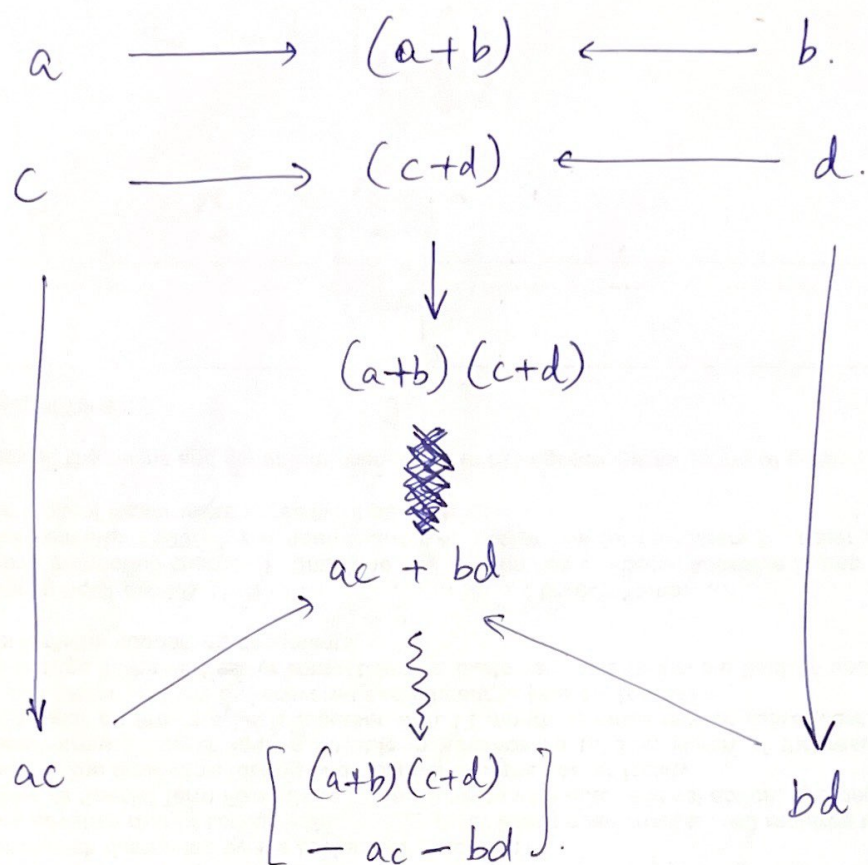
$$ad + bc = (a+b)(c+d) - ac - bd.$$

~~$$10^4 ac + 10^2[(a+b)(c+d) - ac - bd] + bd.$$~~

$$10^4 ac + 10^2[(a+b)(c+d) - ac - bd] + bd.$$



3 multiplications (7).



$$10^4 ac + 10^2 \left[\begin{array}{c} (a+b)(c+d) \\ -ac - bd \end{array} \right] + bd$$

Analysis

$$\begin{aligned}
 T(n) &= 3 \text{ [scribble]} T\left(\frac{n}{2}\right) \\
 &= 3 \times 3 T\left(\frac{n}{4}\right)
 \end{aligned}$$

Facts about logarithms

- ① If $n = 2^k$, then $k = \log_2 n$.
- ② $\log_2 3 \leq 1.585$.
- ③ $a^{\log b} = b^{\log a}$. (8)

$$T(n) = 3 \times 3 \times 3 T\left(\frac{n}{8}\right)$$

$$= 3^k T\left(\frac{n}{2^k}\right)$$

When is $\frac{n}{2^k} = 1$?

$$n = 2^k, \text{ or } k = \log_2 n.$$

$$= 3^{\log_2 n} \times T(1)$$

$$= n^{\log_2 3} \leq n^{1.585}$$

Karatsuba's algorithm [1960]: $O(n^{1.585})$

Conventional algorithm: $O(n^2)$

~~Best algorithm~~ (FFT): $O(n \log_2 n)$

Galactic algorithm [1968]: $O(n \log \log n)$

[2014]: $O(n \log n)$

← Conventional

$n \times n$

← faster

$n \times n^{0.585}$

← faster

$n \times \log n \log n$

← fastest?

$n \times \log_2 n$

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